

MARIA ALM, Austria

WMO: 111370

Lat: 47.406N

Lon: 12.898E

Elev: 2606

StdP: 13.36

Time Zone: 1.00 (EUC)

Period: 05-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	10.2	2.2	6.8	6.4	6.9	8.6	10.5	13.1	37.0	11.3	36.1	3.1	120	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	84.1	66.5	80.7	65.0	77.2	63.7	68.1	80.2	66.5	77.4	65.0	74.7	4.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	99.3	72.7	62.7	93.9	70.7	61.3	89.2	69.1	33.9	79.9	32.6	77.7	31.3	75.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	9.8	8.2	3.1	90.3	5.9	1.7	-1.2	91.6	-4.6	92.6	-8.0	93.5	-12.3	94.8
			2.1	71.6	5.7	1.8	-2.0	72.9	-5.3	74.0	-8.5	75.0	-12.6	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	25.7	29.6	38.2	47.3	54.5	60.7	64.2	63.0	55.0	46.8	36.0	27.8
	DBStd	15.07	7.87	7.65	6.81	6.68	6.67	7.08	6.29	5.94	6.17	6.44	7.40	6.71
	HDD50	3137	754	571	368	127	35	8	0	1	24	138	423	688
	HDD65	7152	1219	991	832	531	328	161	92	110	302	563	870	1153
	CDD50	1612	1	0	1	45	174	331	440	404	173	40	3	0
	CDD65	152	0	0	0	0	3	33	66	48	2	0	0	0
	CDH74	1473	0	0	0	6	88	365	581	405	28	0	0	0
	CDH80	351	0	0	0	1	12	93	152	92	1	0	0	0
Wind	WSAvg	3.4	3.0	3.2	3.8	4.1	3.9	3.7	3.6	3.2	3.0	3.1	3.0	2.9
Precipitation	PrecAvg	55.9	3.0	3.0	3.9	3.5	5.4	6.9	7.6	7.2	5.2	3.9	3.5	3.2
	PrecMax	63.1	7.3	7.9	8.9	6.0	7.9	10.7	11.5	11.3	9.6	6.8	7.2	9.1
	PrecMin	46.5	0.3	0.7	1.5	0.6	1.3	4.1	3.4	4.6	2.1	0.9	0.2	0.8
	PrecStd	4.4	1.8	1.8	2.0	1.6	1.7	1.7	2.0	1.7	1.8	1.6	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	49.0	63.4	73.7	81.2	87.9	88.5	86.5	77.8	67.1	61.3	50.2
		MCWB	45.4	40.5	48.8	54.7	61.8	66.8	69.0	68.2	64.7	58.0	49.6	41.3
	2%	DB	41.1	44.2	58.0	68.9	76.0	82.4	84.2	82.5	72.7	62.9	55.9	41.8
		MCWB	39.0	38.5	46.1	53.2	59.4	64.8	66.6	66.9	62.4	55.2	48.0	39.9
	5%	DB	36.5	40.9	53.5	65.1	71.8	78.2	80.6	78.6	68.9	59.7	49.1	37.8
		MCWB	34.9	36.7	43.9	51.6	57.5	63.1	65.8	65.4	60.2	53.3	45.4	36.3
	10%	DB	34.7	38.5	48.8	61.0	67.8	73.8	76.6	74.4	65.3	56.5	45.3	35.7
		MCWB	33.4	35.2	41.4	49.6	55.5	61.4	64.1	63.6	58.2	51.9	43.3	34.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	41.1	49.8	56.1	63.7	69.0	70.4	70.5	65.7	59.1	50.9	42.7
		MCDB	50.4	47.4	61.9	71.1	76.6	83.0	84.8	83.5	74.9	64.9	58.8	46.9
	2%	WB	38.4	38.9	46.7	53.9	61.1	66.4	68.1	68.0	63.3	56.2	47.9	39.6
		MCDB	41.0	43.4	56.7	67.1	73.1	79.1	80.6	79.0	71.5	61.3	54.6	41.7
	5%	WB	35.2	37.1	44.4	52.0	58.8	64.6	66.6	66.2	61.1	54.2	45.6	36.7
		MCDB	36.4	40.3	52.3	63.4	69.6	75.2	77.7	76.1	67.2	58.5	49.2	37.9
	10%	WB	33.6	35.6	42.0	50.1	56.5	62.6	65.0	64.4	59.0	52.3	43.3	34.6
		MCDB	34.5	37.8	48.2	59.8	65.8	71.2	74.5	73.3	64.1	56.0	45.3	35.4
Mean Daily Temperature Range	5% DB	MDBR	10.2	13.1	16.9	20.2	20.3	19.1	19.4	19.3	16.5	15.6	10.8	8.5
		MCDBR	10.8	15.9	25.6	29.1	29.3	28.9	27.7	27.0	22.4	19.7	17.8	10.2
		MCWBR	8.7	10.6	14.4	14.6	13.6	12.2	12.0	12.5	12.5	12.4	11.6	7.9
	5% WB	MCDBR	10.5	13.9	24.5	27.3	26.9	25.2	24.9	24.8	21.2	18.2	16.4	9.5
		MCWBR	8.6	9.6	13.9	14.0	12.9	11.0	11.5	12.4	12.3	11.7	10.8	7.5
Clear-Sky Solar Irradiance	taub	0.277	0.306	0.350	0.391	0.405	0.412	0.411	0.404	0.369	0.339	0.307	0.277	
	taud	2.382	2.298	2.249	2.232	2.252	2.277	2.280	2.320	2.393	2.450	2.470	2.440	
	Ebn at Noon	258	271	274	273	273	271	269	266	266	257	245	245	
	Edh at Noon	22	30	36	41	42	41	40	37	32	26	20	19	
All-Sky Solar Radiation	RadAvg	413	709	1020	1301	1485	1643	1672	1488	1088	762	429	336	
	RadStd	50	89	121	164	179	172	162	178	145	81	71	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)	+0.74	N/A	N/A	N/A	+0.70	+0.68	N/A	N/A	+112	+50

Nomenclature: See separate page